

LAB SESSION 13

Programming Exercise:

Question 1: Overload the * operator to multiply two Matrix objects (define a simple Matrix class with a 2D array).

```
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o  
tut1 } ; if ($?) { .\tut1 }  
Enter elements of Matrix A:  
Enter 9 elements:  
1 2 3  
4 5 6  
7 8 9  
Enter elements of Matrix B:  
Enter 9 elements:  
9 8 7  
6 5 4  
3 2 1  
  
Result of A * B:  
30 24 18  
84 69 54  
138 114 90  
PS C:\C++ PROGRAMS> █
```

```

#include <iostream>
using namespace std;

const int SIZE = 3; // Fixed size for simplicity

// Matrix class
class Matrix {
private:
    int mat[SIZE][SIZE];

public:
    // Default constructor initializes with zeros
    Matrix() {
        for (int i = 0; i < SIZE; i++)
            for (int j = 0; j < SIZE; j++)
                mat[i][j] = 0;
    }

    // Function to input matrix elements
    void inputMatrix() {
        cout << "Enter " << SIZE * SIZE << " elements:" << endl;
        for (int i = 0; i < SIZE; i++)
            for (int j = 0; j < SIZE; j++)
                cin >> mat[i][j];
    }

    // Function to display matrix
    void displayMatrix() const {
        for (int i = 0; i < SIZE; i++) {
            for (int j = 0; j < SIZE; j++)
                cout << mat[i][j] << " ";
            cout << endl;
        }
    }

    // Overload * operator for matrix multiplication
    Matrix operator*(const Matrix &other) {
        Matrix result;
        for (int i = 0; i < SIZE; i++) {
            for (int j = 0; j < SIZE; j++) {
                for (int k = 0; k < SIZE; k++) {
                    result.mat[i][j] += mat[i][k] * other.mat[k]
[j];
                }
            }
        }
        return result;
    }
};

// Main function
int main() {
    Matrix A, B, C;

    cout << "Enter elements of Matrix A:\n";
    A.inputMatrix();

    cout << "Enter elements of Matrix B:\n";
    B.inputMatrix();

    C = A * B; // Using overloaded * operator

    cout << "\nResult of A * B:\n";
    C.displayMatrix();

    return 0;
}

```